When $x = -2$, $f''(-2) = 12(-2) + 6 = -18 < 0$

$\therefore f(x)$ attains local maximum at $x = -2$ and local maximum value is obtained from (1) by substituting the value $x = -2$.

$f(-2) = 2(-2)^3 + 3(-2)^2 - 12(-2)$

$= -16 + 12 + 24 = 20.$

When $x = 1$, $f''(1) = 12(1) + 6 = 18$

$f(x)$ attains local minimum at $x = 1$ and the local minimum value is obtained by substituting $x = 1$ in (1).

$f(1) = 2(1) + 3(1) - 12(1) = -7$

Extremum values are -7 and 20.

Example 6.26

Find the absolute (global) maximum and absolute minimum of the function

$f(x) = 3x^5 - 25x^3 + 60x + 1$ in the interval $[-2, 1]$

Solution :

$f(x) = 3x^5 - 25x^3 + 60x + 1$ ... (1)

$f'(x) = 15x^4 - 75x^2 + 60 = 15(x^4 - 5x^2 + 4)$

$f'(x) = 0 \Rightarrow 15(x^4 - 5x^2 + 4) = 0$

$\Rightarrow (x^2 - 4)(x^2 - 1) = 0$

$\Rightarrow x = \pm 2$ (or) $x = \pm 1$

Of these four points $-2, \pm 1 \in [-2, 1]$ and $2 \notin [-2, 1]$

From (1)

$f(-2) = 3(-2)^5 - 25(-2)^3 + 60(-2) + 1 = -15$

When $x = 1$

$f(1) = 3(1)^5 - 25(1)^3 + 60(1) + 1 = 39$

When $x = -1$

$f(-1) = 3(-1)^5 - 25(-1)^3 + 60(-1) + 1 = -37$

Absolute maximum is 39 and

Absolute minimum is -37.

6.3 Applications of Maxima and Minima

6.3.1 Problems on profit maximization and minimization of cost function:

Example 6.27

For a particular process, the cost function is given by $C = 56 - 8x + x^2$, where C is cost per unit and x, the number of unit's produced. Find the minimum value of the cost and the corresponding number of units to be produced.

Solution:

$C = 56 - 8x + x^2$

Differentiate with respect to x,

$\frac{dC}{dx} = -8 + 2x$ and $\frac{d^2C}{dx^2} = 2$

$\frac{dC}{dx} = 0 \Rightarrow -8 + 2x = 0 \Rightarrow x = 4$

When $x = 4$, $\frac{d^2C}{dx^2} = 2 > 0$

$\therefore C$ is minimum when $x = 4$

The minimum value of cost = $56 - 32 + 16 = 40$

The corresponding number of units produced = 4.

Example 6.28

The total cost function of a firm is $C(x) = \frac{x^3}{3} - 5x^2 + 28x + 10$, where x is the output. A tax at the rate of ₹ 2 per unit of output is imposed and the producer adds it to his cost. If the market demand function is given by $p = 2530 - 5x$, where p is the price per unit of output, find the profit maximizing the output and price.

Solution:

Total revenue: $R = p\ x$

$= (2530 - 5x)x = 2530x - 5x^2$

Tax at the rate ₹ 2 per x unit = $2x$.

06_11th_BM-STAT_Ch-6-EM.indd 144 21-04-2020 12:22:03 PM